

ANIL PRADHAN

BACKEND ENGINEER | AI SYSTEMS BUILDER

Computer Science undergraduate passionate about building scalable backend systems and AI-powered applications. Experienced in FastAPI, LangGraph, RAG pipelines, and cloud deployment. Eager to contribute to impactful products and grow as a software engineer.

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EDUCATION

ITER, SOA University, Bhubaneswar
B.Tech in Computer Science & Engineering
2024 – 2028 | CGPA: 9.11/10

Odisha Board of Secondary Education
Class XII (PCM with IT)
2022 – 2024 | Percentage: 85%

EXPERIENCE

College Ambassador | Techfest, IIT Bombay

Jun 2026 – Present

- Selected as a Campus Ambassador for Techfest, IIT Bombay, participating in technical and industry-oriented initiatives.
- Contributed to web development and marketing by successfully completing multiple project-based assignments.
- Conducted market research, prepared a business strategy presentation, and qualified for Round 2 of the FlowFinance Growth Strategy & Business Development Challenge.

TECHNICAL PROJECTS

AI Career Mentor | Live Demo | Solo Full-Stack Developer

March 2026 – Present

- Multi-Agent Orchestration & Concurrency:** Orchestrated 7 specialized AI agents using a parallel LangGraph DAG; reduced end-to-end latency by 60% (from 4 mins to 60s) by executing resume parsing and market scraping concurrently.
- System Resilience & Fault Tolerance:** Built failover architecture with custom Circuit Breakers and LLM fallback chains (Groq ⇌ Cerebras ⇌ NVIDIA NIM), preventing request drops during API rate-limit spikes.
- Real-time Bidirectional Streaming & State Machines:** Designed a full-duplex WebSocket protocol with Gemini Multimodal Live API for low-latency audio coaching; built a 7-phase FSM coding interview simulator with Monaco IDE.
- High-Efficiency RAG & Semantic Search:** Optimized ChromaDB RAG using local ONNX embeddings (all-MiniLM-L6-v2) to eliminate API costs; built a keyword-search fallback to prevent OOM container crashes.
- Rate-Limiting, Caching & Telemetry:** Integrated Upstash Redis middleware for sliding-window rate limiting and JWT validation; built a telemetry dashboard tracking API latencies and token cost rollups.
- Data Durability, Dockerization & CI/CD:** Enforced data validation with Pydantic v2, persisting entries to Postgres with PgBouncer pooling; deployed multi-stage Docker builds with a 116 Pytest CI/CD suite.

FastAPI LangGraph ChromaDB ONNX Docker Upstash Redis Next.js 14 Gemini Live

Task Pilot-AI | Live Demo | Integration Lead • Team IdeaForg-E

June 2026 – July 2026

- Solved Developer Context Overload:** Aggregated engineering tasks from 7 sources (Jira, GitHub, Slack, Email, Calendar, Meetings, Incidents) into a unified DB, saving developers ~2 hours daily.
- Recovered Untracked "Hidden" Tasks:** Developed an LLM and rule-based extraction system to parse emails and Slack chats for hidden action items, capturing ~35% of previously untracked tasks.
- Reduced LLM Costs and API Failures:** Built local fallback algorithms to reduce token costs and bypass Groq API rate limits; used fuzzy text-matching for deduplication and local Slack parsing.
- Prevented Database Locking and Workflow Freezes:** Configured SQLite in WAL mode to allow concurrent multi-agent writes without locking; built a self-healing monitor to auto-restart stuck pipeline runs.
- Automated Real-Time Emergency Re-Prioritization:** Created a chat command API allowing developers to inject critical P1 incidents; autonomously re-ran the 6-stage pipeline to update schedules in under 15s.

FastAPI LangGraph SQLite LLMs React Redis

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL
Backend: FastAPI, LangGraph, Pydantic, SQLAlchemy 2.0, WebSocket, Prometheus, Sentry, pdfplumber, edge-tts
Databases: Neon Postgres, SQLite, Upstash Redis, ChromaDB (Vector Store)
DevOps & Infra: Docker, GitHub Actions (CI/CD), Vercel, Render.com

Frontend: HTML, CSS, Tailwind CSS, Axios, React
AI / LLM & RAG: Groq Cloud, Cerebras Cloud, NVIDIA NIM, Gemini Live API, Tavily, Serper.dev, DuckDuckGo
Auth & Security: JWT (python-jose), Google OAuth 2.0, bcrypt, CORS Middleware, Rate Limiting, HTTPS
Tools: VS Code, Git, Swagger UI, Cline, GitHub Copilot, Gemini Code Assist, OpenAI Codex, AntiGravity IDE

ACHIEVEMENTS & CERTIFICATIONS

- Google Cloud Arcade Program (2024–25): Completed GCP challenges, earning 75 Arcade Points.
- Solved 200+ problems across GFG and LeetCode, awarded GFG Certificate and T-shirt.
- Certifications: Credentials in Python, AI, and Web Dev from IBM, Cisco, Anthropic, Infosys, Udemy.
- Competitions: Participated in multiple IIT/NIT coding contests and hackathons.
- Hackathons: Built production-grade MVP solutions for Microsoft AI DevDays (\$80K+) and Amazon Nova AI (\$95K+).
- Active participant in Developer Communities (GDG Events) and college tech events.